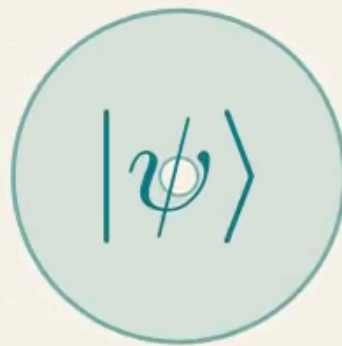




Day 3

Intro to Quantum Computing

KAUST ACADEMY + KGSP

Yesterday: teleportation — every count was 4.

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Today: why is quantum **fast**?

Pop-science says

“A quantum computer tries all answers at once.”

Wrong — it measures **one** outcome.

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“A quantum computer tries all answers at once.”

Wrong — it measures **one** outcome.

What's actually true

*Amplitudes for wrong answers cancel.
The right answer's amplitude survives.*

“Quantum gates progressively manipulate the state vector until it points almost entirely at the answer you care about.”

— 3Blue1Brown, “But what is a quantum computer?”

Today: **three algorithms**, one engine.

The engine — interference: how wrong answers erase themselves.

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Problems first, names after — each algorithm will earn its name.

Interference — the engine

You already own a device that runs on interference.



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They add a wave **180° out of phase** with the noise — where they meet, it cancels.

add them where they meet $\rightarrow$ flat line

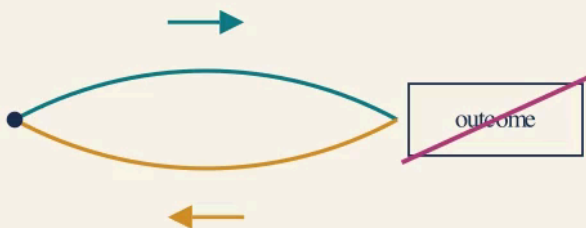
anti-noise



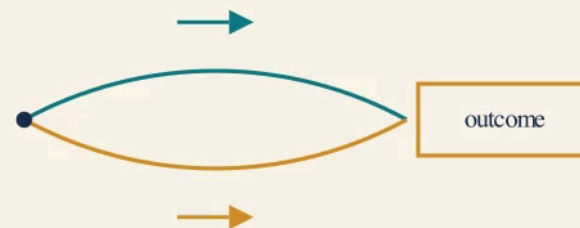
noise

silence — the waves cancel

two paths into the same outcome — add their arrows tip-to-tail

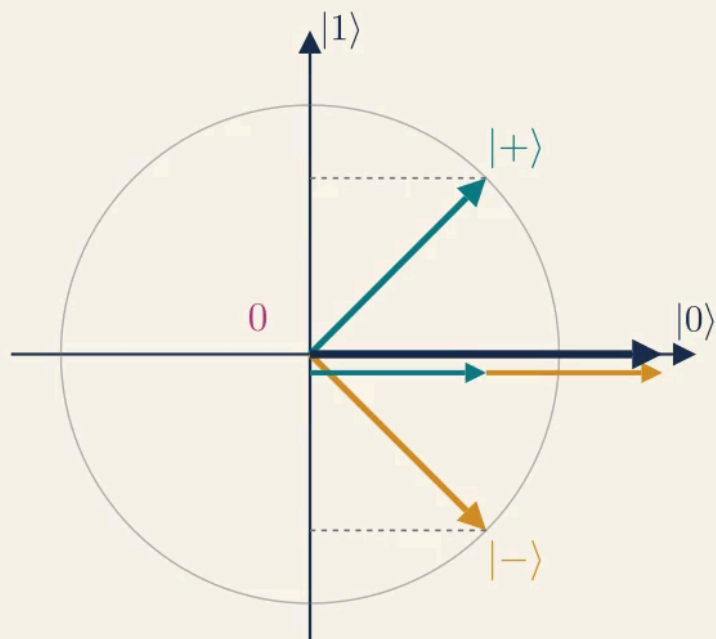


sum=0 — this outcome never happens



sum doubles — this outcome dominates

components add — arrows on the Bloch sphere don't



$$|+\rangle + |-\rangle = ?$$

$$|1\rangle : \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$$

$$|0\rangle : \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$|+\rangle + |-\rangle = \sqrt{2} |0\rangle \propto |0\rangle$$

Two paths lead to the same outcome.
Their amplitudes **add first, then** you square.

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A **sign** decides whether an outcome exists. Remember $|-\rangle$'s minus from yesterday? That dial.

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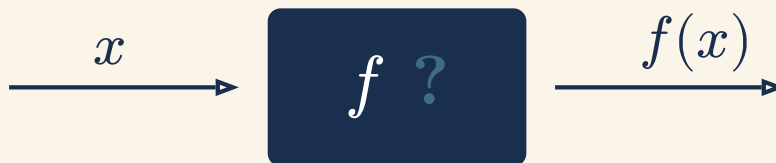
An algorithm is a **choreography of signs**.
Design it well and the wrong answers erase themselves.

*Quantum speedup is amplitudes cancelling — nothing more magical than
headphones.*

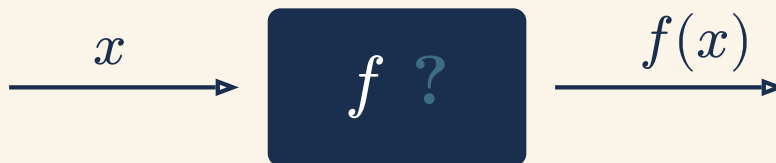
A first algorithm

One global question, answered in a single query.

A black-box function $f(x)$. You may call it **once**.



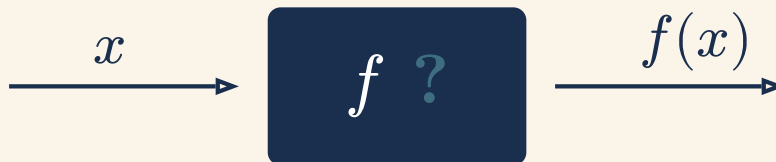
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It's either **constant** (same output for every input)
or **balanced** (0 on half the inputs, 1 on the other half).

Which kind is it?

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A function you may only **ask** — never open up — is called an **oracle**.

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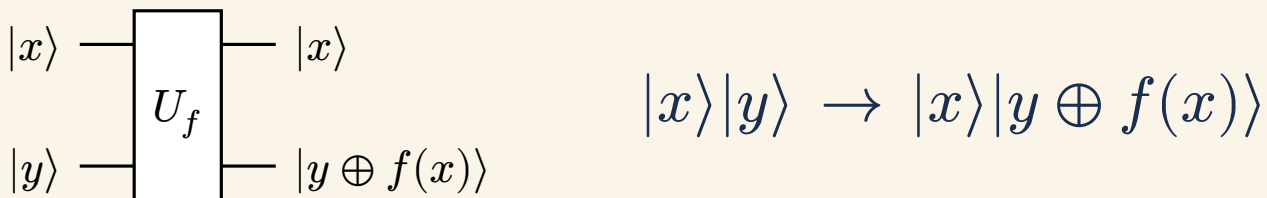
Quantum: one call.

How can a single query decide a **global** property?

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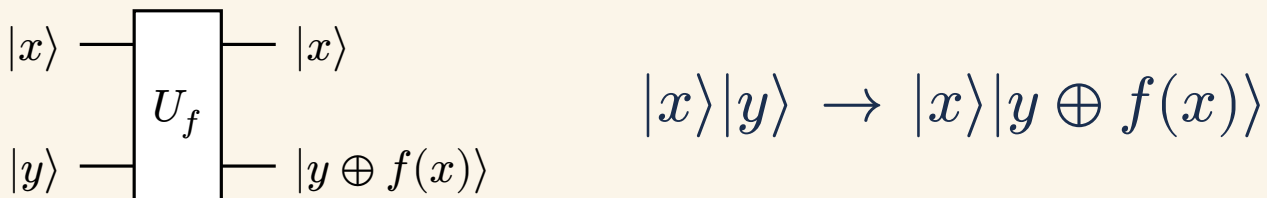
So the oracle is handed to the machine as a **reversible gate** that writes $f(x)$ onto a **helper** qubit:



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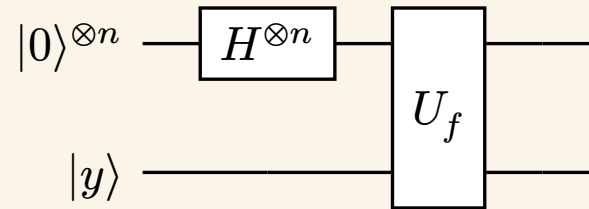


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The oracle, packaged as a gate, is a **quantum oracle**.

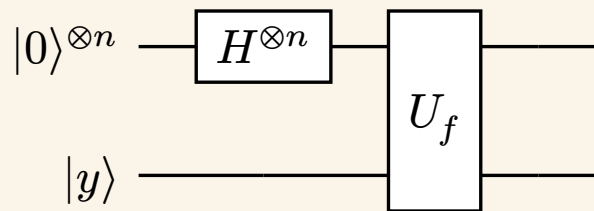
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One call to the quantum oracle now writes $f(x)$ onto **every branch** of the superposition.

But it's trapped: reading **any** helper collapses everything to one random x .

We need $f(x)$ as a **sign**, not a bit — signs can interfere.

The trick: prepare the helper in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

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The enabling feature: $X|-\rangle = -|-\rangle$ — the flip can only **rescale** $|-\rangle$ (it's an **eigenstate**), and that scalar -1 belongs to the whole branch that triggered it.

The whole algorithm, in three moves:

1. Hadamard every input $\rightarrow$ an equal superposition of all inputs.

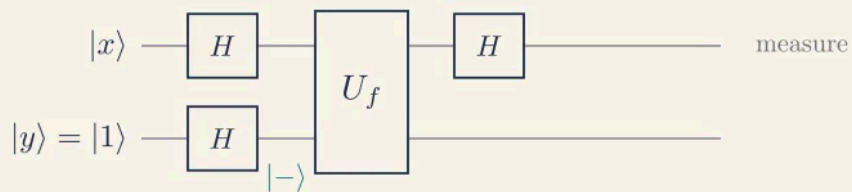
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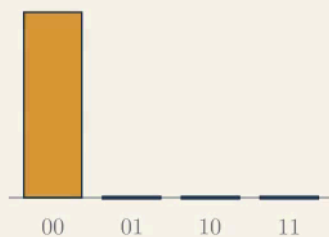
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3. Hadamard again $\rightarrow$ the signed branches **interfere**.

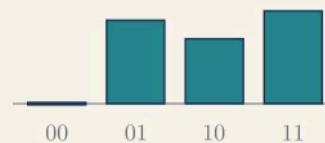
one query, one measurement $\rightarrow$ constant or balanced



$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$



constant $\rightarrow$ all on $|00\rangle$



balanced $\rightarrow |00\rangle$ empty

Measure. All zeros $\rightarrow$ **constant**. Anything else $\rightarrow$ **balanced**.

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*One quantum query resolves what classical needs many for —
because interference decides the answer, not the value.*

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Including the next one: a search actually worth doing.

The needle in a haystack

N items, one marked.

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Anywhere you can check an answer fast, this search finds it — no structure needed.

That “easy to **check**, hard to **find**” shape is exactly the **NP** family of problems — the hardest of them are the NP-hard ones.

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The quantum oracle is **phase kickback again** — the Deutsch–Jozsa stamp, with the oracle now asking “*is this a marked item?*”:

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The mark is **invisible** — every probability is still $1/N$. The sign has to be converted into something measurable.

all eight identical — the mark is invisible until it interferes



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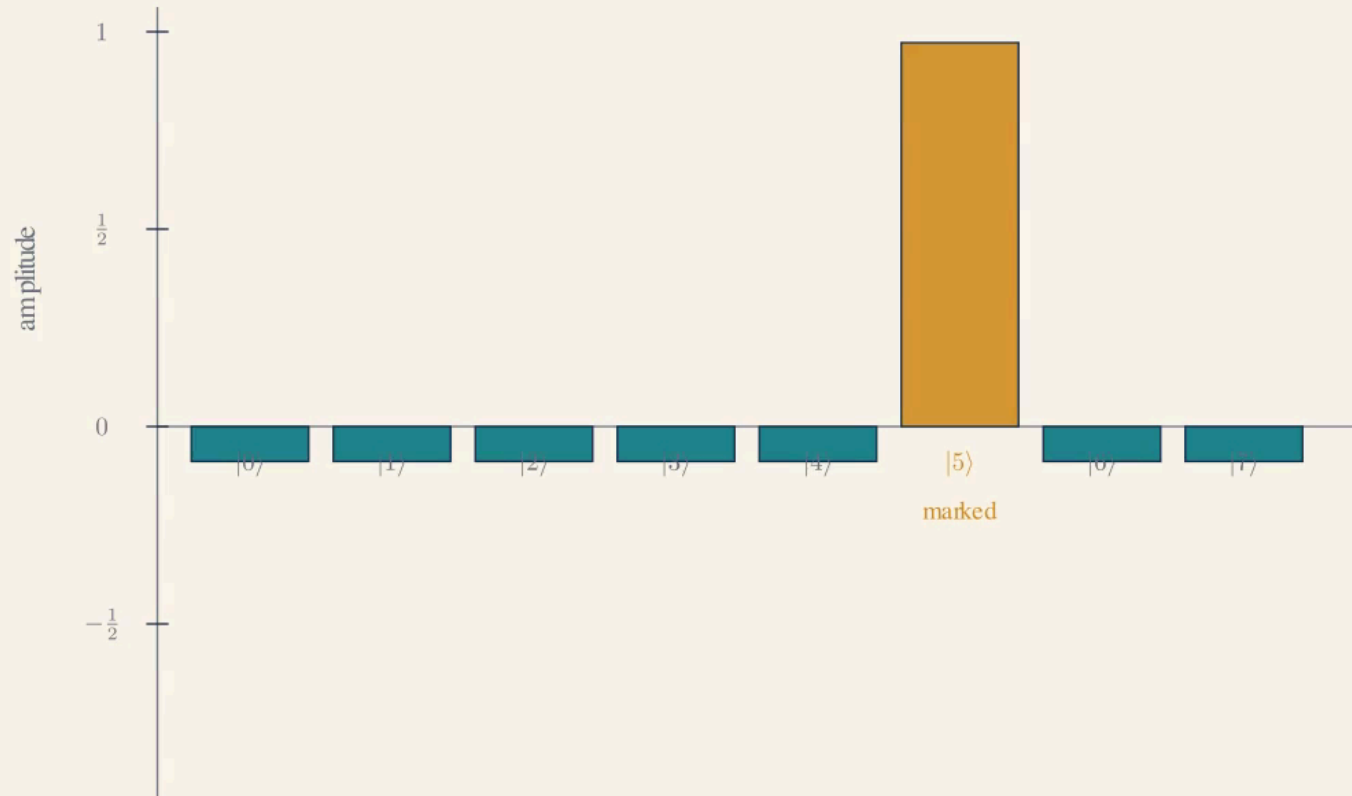
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This spread-and-rebound move is called **diffusion**.

2 iterations $\rightarrow$ the marked amplitude dominates



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Start — uniform: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

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Diffusion $a \rightarrow 2m - a$: $(0, 0, 1, 0)$

Marked amplitude = 1, so $P = 1^2 = 100\%$ — after **one** iteration.

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Predict: run one more iteration anyway.

Better, the same — or worse? (show of hands)

Oracle flips item 3 again: $(0, 0, -1, 0)$

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Mean $m = -\frac{1}{4}$; diffusion: $(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$

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The search **overshoots** — more is not better. (You'll do both rounds on paper today.)

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And it's **provably optimal**: for unstructured search, **no** quantum algorithm can beat $\sqrt{N}$.

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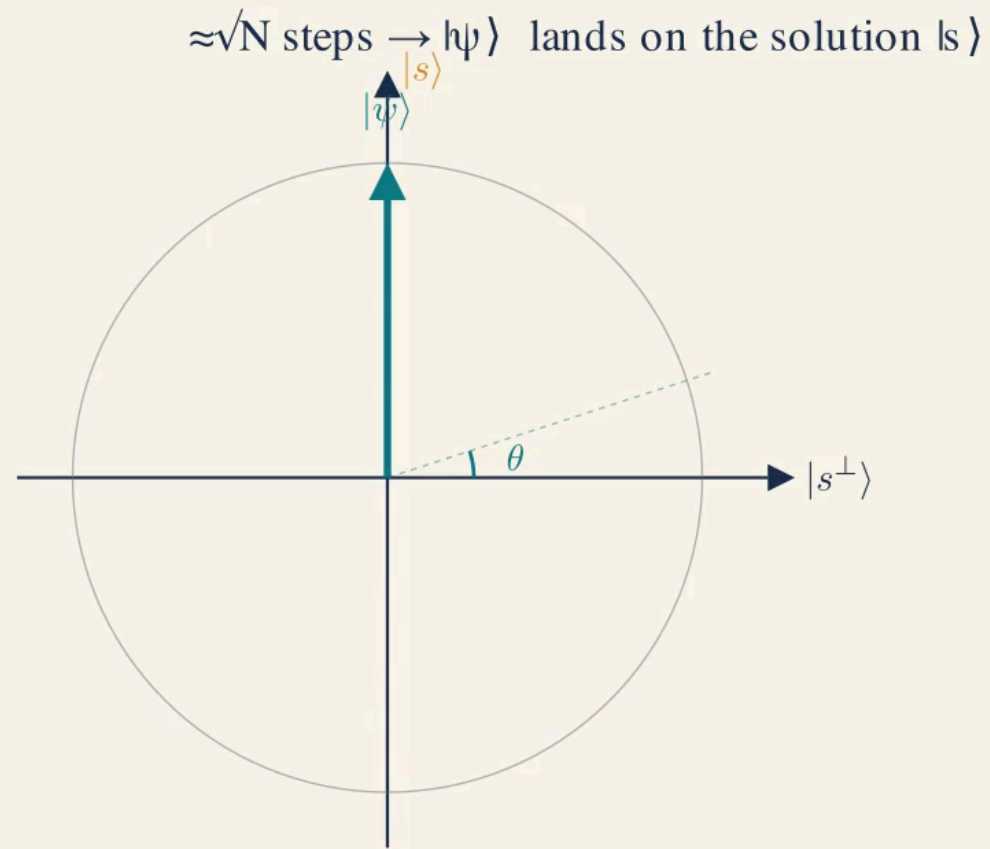
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$\approx \sqrt{N}$ small turns sweep the 90° to $|s\rangle$ — and keep turning **past** it. Overshoot, explained.



Each step rotates by 2θ ; the target is 90° away. Iterations needed:

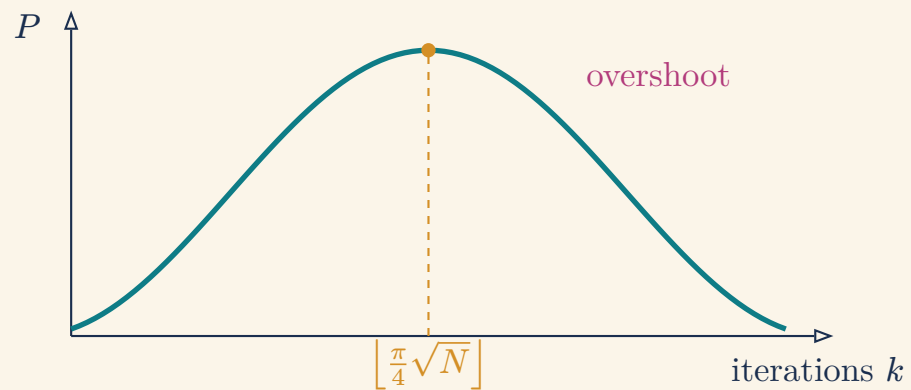
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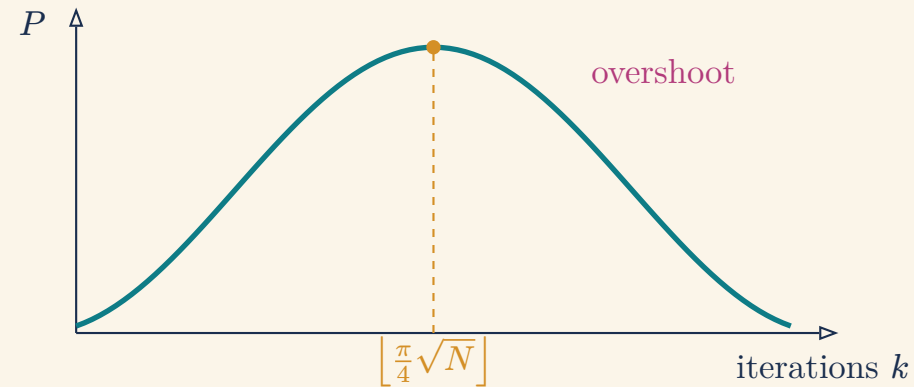
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For $N = 4$: $\left\lfloor \frac{\pi}{4} \cdot 2 \right\rfloor = 1$ — exactly the one iteration that hit 100%.

Success probability vs. iterations — rise, **peak**, fall:



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Stop at the peak. Keep rotating and you swing past $|s\rangle$ — the odds fall.

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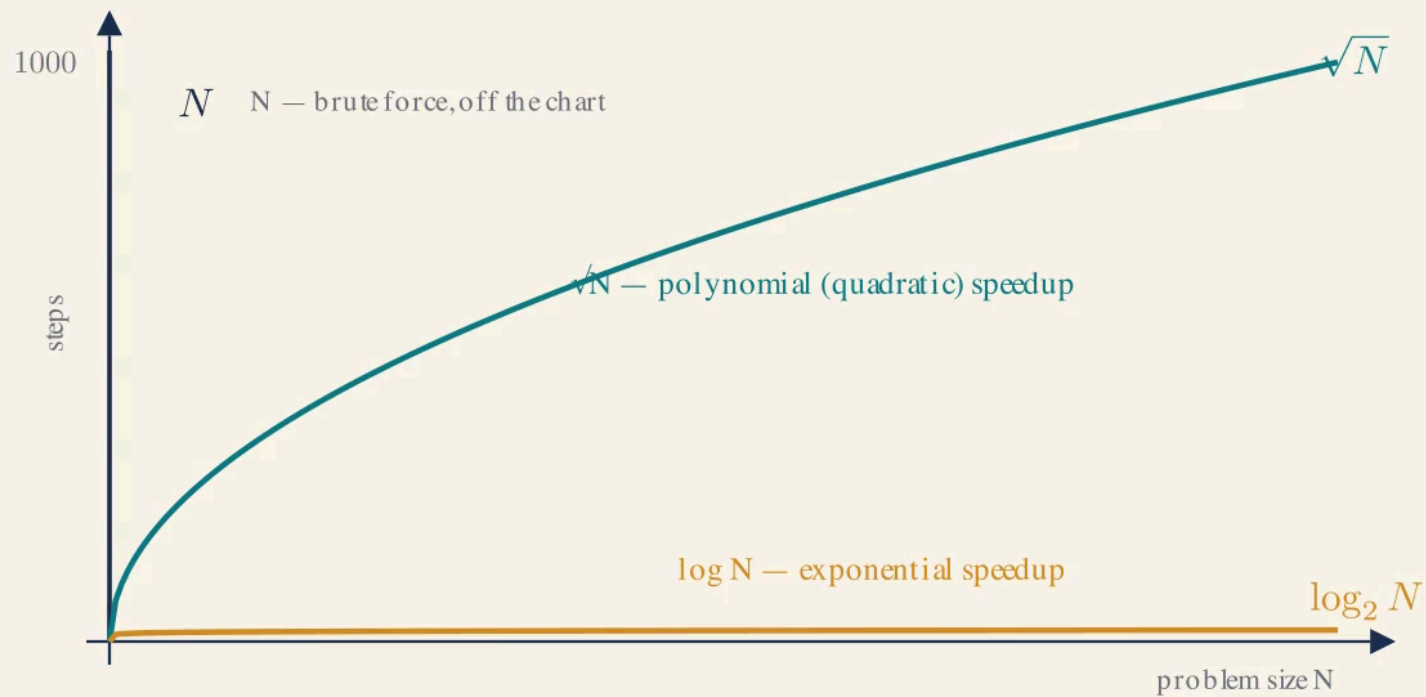
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A trillion items: $10^{12} \rightarrow 10^6$. A million times fewer steps — real, but not “instant.”

For an **exponential** win, the problem needs **structure** to exploit...

zoom into the floor $\times 1000$ — now you can see the classes



Grover generalises to **any graph**.

[Small graph — 6–8 nodes — with a walker on it]

Search-on-a-graph via a quantum walk · the lab's aspirational tier.

The lock on the internet

One more algorithm — the one that made governments pay attention.

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That one-way gap **is** a lock — it's **RSA**, the padlock behind https, banking, messaging.

Everyone can lock (multiply); only the key-holder can unlock (factor).

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No circuit today — the shape is the point. (More on Day 5.)

How big is the gap, for the numbers the internet actually uses?

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Best classical

$\sim 10^{14}$ years

$\approx 20,000 \times$ the age of the universe
(1.4×10^{10} years)

Shor, on a big machine

~ 8 hours

2019 estimate, a fault-tolerant
machine of 20 million qubits

Estimates vary wildly — the **gap** is the point, not the numbers.

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The point today is what Shor threatens, and why. (Day 5: where the hardware actually is.)

Today, in four lines:

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Shor — structure buys an exponential win; RSA's clock is ticking.

After lunch you'll code Grover.

Baseline: find one marked item in 4.

Stretch: 16 items — tune the iteration count, watch it peak.

Aspirational: a quantum walk on a small graph.

After lunch

- ✓ Open today's lab on MOLAB (link posted)
- ✓ Give it a moment — it auto-installs Qiskit
- ✓ Local simulator by default; IBM account only for real hardware

Snagged? grab a TA.
Setup stays mechanical.